

Chapter 9: The Number Plane

Three-week revision booklet · 6 worksheets · 114 questions

NSW syllabus outcome

MA4-LIN-C-01 — creates and displays number patterns, and graphs linear relationships · MAO-WM-01

NAME _____

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How to use this booklet

There are six worksheets, two for each week, and each one runs across **two pages**. Most questions come with a real number plane to plot on, or a table of values to complete, so you are working the way you will in the topic test. Every sheet is split into **Easy**, **Medium** and **Challenging**, and each section opens with a short hint box telling you how to tackle that kind of question. Ruled space is only given where a question needs working out — if the answer belongs in the table or on the plane, write it there. The Challenging questions are written in words only: read them carefully and **draw your own diagram** in the space provided. The final worksheet mixes every skill together, so it is the best check that you are ready for the topic test.

Week	Worksheet	Skill
1	1	Location using coordinates
1	2	The number plane
2	3	Quadrants of the number plane
2	4	Graphing tables of values
3	5	Completing tables of values
3	6	Graphing linear equations

Progress tracker

Worksheet	Date completed	Score / 18	Checked by
1 Location using coordinates			
2 The number plane			
3 Quadrants of the number plane			
4 Graphing tables of values			
5 Completing tables of values			
6 Graphing linear equations			

Worksheet 1: Location using coordinates

Week 1 · Worksheet 1 of 6 · 18 questions

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A coordinate tells you where something is. Read across first, then up.

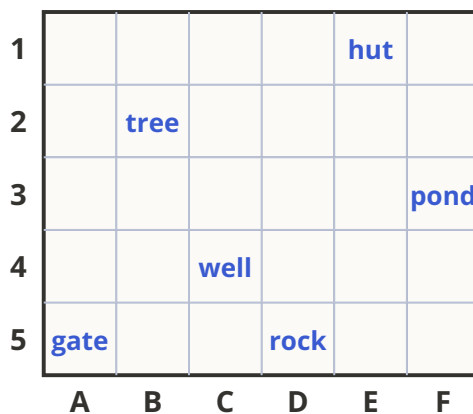
EASY

9 QUESTIONS

READING A LOCATION

- On a map grid, give the **column letter first**, then the row number: B3.
- On a number plane, read **across** the x-axis first, then **up** the y-axis.
- Either way it is always across before up.

Write the grid reference for each thing on the map.



1. The tree

Answer: _____

2. The hut

Answer: _____

3. The well

Answer: _____

4. The gate

Answer: _____

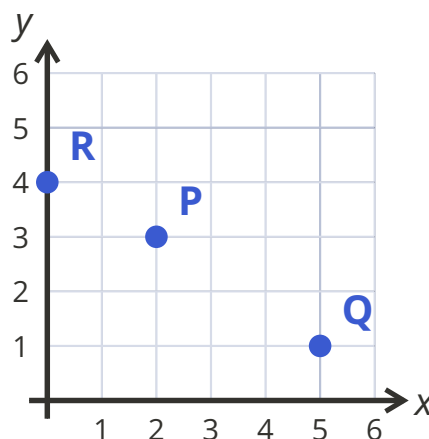
5. The pond

Answer: _____

6. The rock

Answer: _____

Write the coordinates of each point on the number plane.



7. Point P

Answer: _____

8. Point Q

Answer: _____

9. Point R

Answer: _____

Worksheet 1: Location using coordinates

Week 1 · Worksheet 1 of 6 · continued

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MEDIUM

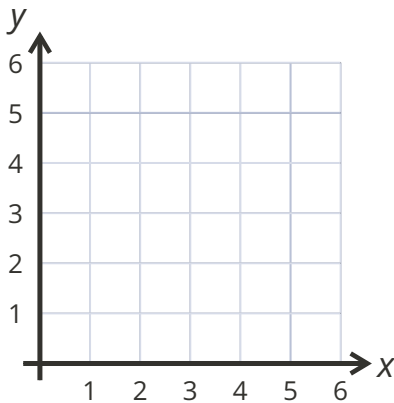
6 QUESTIONS

PLOTTING A POINT

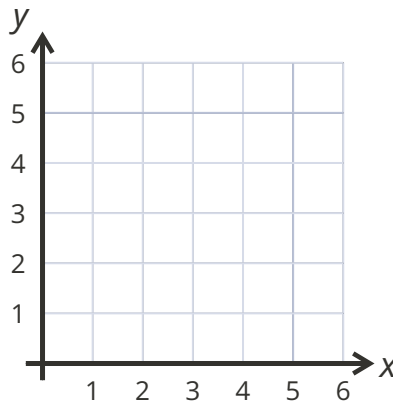
- Start where the axes cross, count across, then count up.
- Mark the point with a neat dot and write its letter beside it.

Plot and label the points on each number plane.

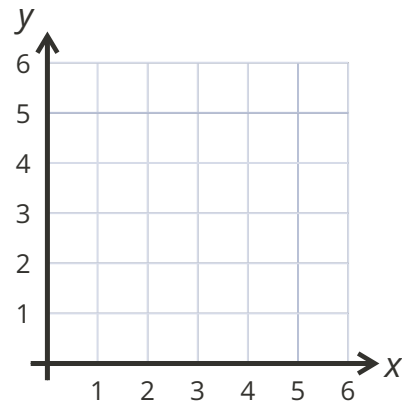
10. A(1, 4), B(3, 0) and C(5, 5)



11. D(0, 2), E(4, 3) and F(6, 6)



12. G(2, 2), H(5, 4) and J(3, 6)



Answer each question.

13. A point is 4 across and 2 up. Write its coordinates.

Answer: _____

14. Write the coordinates of the point where the two axes cross.

Answer: _____

15. On a map grid, is C4 the same square as 4C? Explain.

Answer: _____

CHALLENGING

3 QUESTIONS

BEFORE YOU START

- Draw and label your axes first, then plot every point before you answer.

Read each one, draw your own diagram, then answer the question.

16. Draw a number plane with both axes going from 0 to 6. Plot A(1, 1), B(5, 1) and C(5, 4), then join them up and name the shape.

Draw your diagram here

Answer: _____

17. Draw a map grid with columns A to E and rows 1 to 4. Put a star at C2 and a circle at E4, then describe how to get from one to the other.

Draw your diagram here

Answer: _____

18. Explain why the point (2, 5) is in a different place from the point (5, 2). Use a sketch of a number plane.

Draw your diagram here

Answer: _____

Worksheet 2: The number plane

Week 1 · Worksheet 2 of 6 · 18 questions

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The number plane has an x-axis across and a y-axis up. A point is written (x, y).

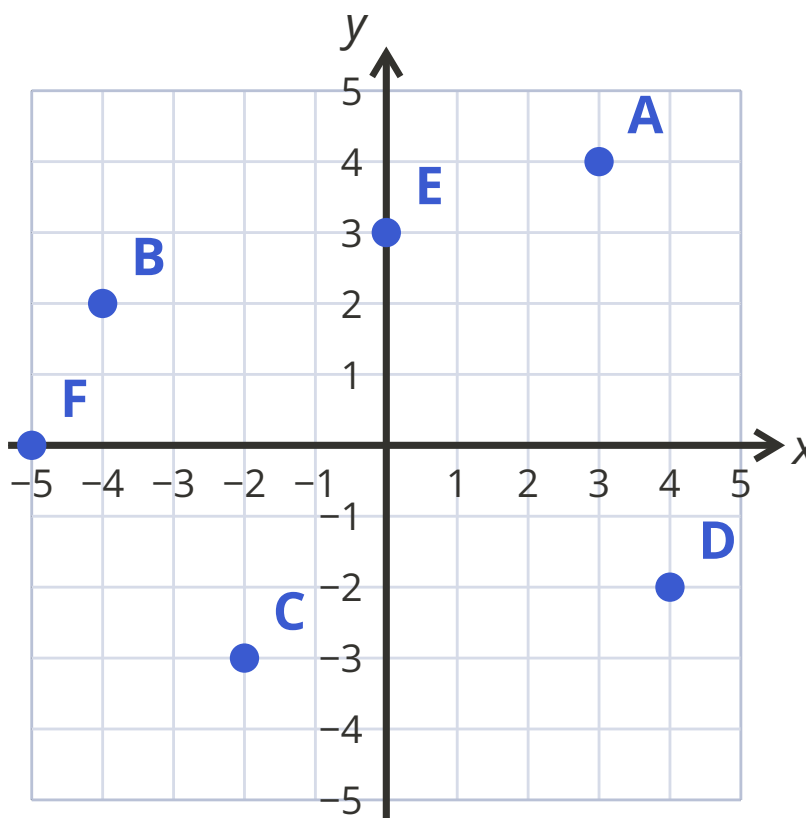
EASY

9 QUESTIONS

READING A POINT

- Across the x-axis **first**, then up or down the y-axis.
- Left of the middle and below it, the numbers are **negative**.
- A point on an axis has a **0** in it: (0, 3) or (-5, 0).

Use the number plane to write the coordinates of each point.



1. Point A

Answer: _____

2. Point B

Answer: _____

3. Point C

Answer: _____

4. Point D

Answer: _____

5. Point E

Answer: _____

6. Point F

Answer: _____

Answer each question.

7. Which axis do all points like (0, 4) lie on?

Answer: _____

8. Which axis do all points like (-3, 0) lie on?

Answer: _____

9. Write the coordinates of the point where the axes cross.

Answer: _____

Worksheet 2: The number plane

Week 1 · Worksheet 2 of 6 · continued

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MEDIUM

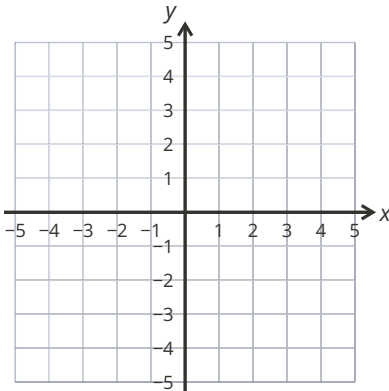
6 QUESTIONS

PLOTTING AND MOVING POINTS

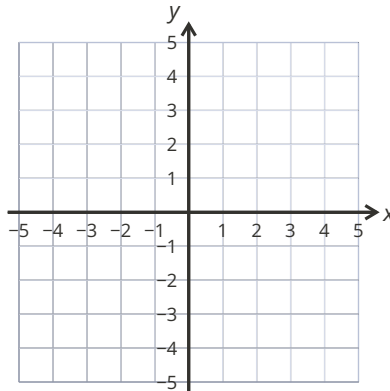
- Count across first, then up or down, and mark the point with a neat dot.
- Moving right adds to x , moving left subtracts. Moving up adds to y .

Plot and label the points on each number plane.

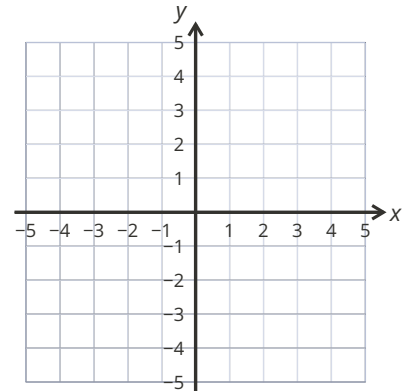
10. A(2, 3), B(-3, 4) and C(-2, -3)



11. P(0, 4), Q(-5, 0) and R(3, -4)



12. D(-1, -5), E(5, 2) and F(-4, -1)



Answer each question.

13. A point is 3 right and 5 down from where the axes cross. Write it.

Answer: _____

14. A(2, 5) is moved 4 left. Write the coordinates of the new point.

Answer: _____

15. P(-2, 1) and Q(4, 1) are the ends of a horizontal line. Find its length.

Answer: _____

CHALLENGING

3 QUESTIONS

BEFORE YOU START

- Plot the points first. For a horizontal or vertical line, the length is the difference between the coordinates that change.

Read each one, draw your own diagram, then answer the question.

16. A(1, 2), B(5, 2) and C(5, 6) are three corners of a rectangle. Plot them, find the fourth corner, then work out the area.

Draw your diagram here

17. R(3, -2) and S(3, 6) are the ends of a vertical line. Plot them and find the length of RS.

Draw your diagram here

18. Explain why (3, 5) and (5, 3) are different points. Use a sketch of a number plane to show the difference.

Draw your diagram here

Answer: _____

Answer: _____

Answer: _____

Worksheet 3: Quadrants of the number plane

Week 2 · Worksheet 3 of 6 · 18 questions

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The two axes cut the plane into four quadrants, numbered anticlockwise from the top right.

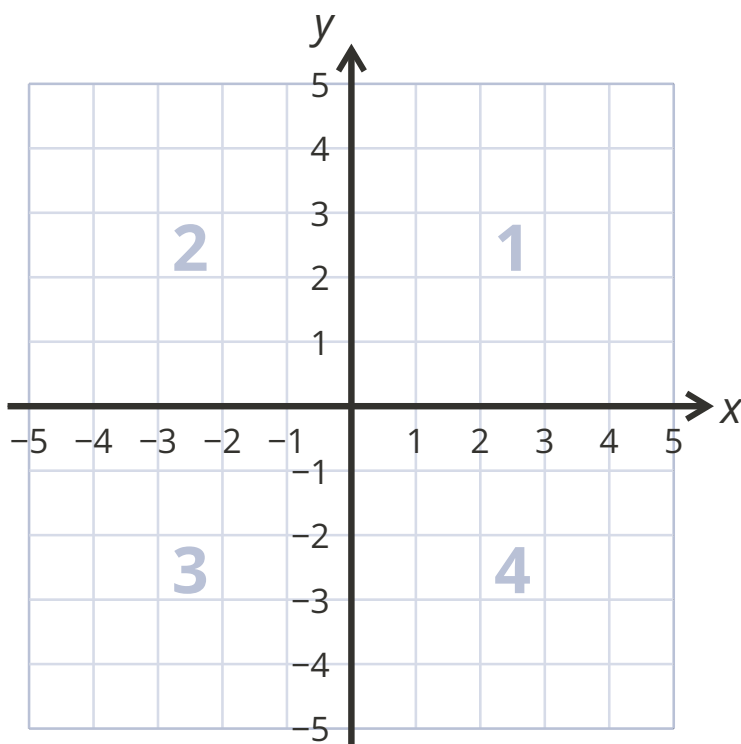
● EASY

9 QUESTIONS

NAMING THE QUADRANT

- Quadrant 1 is (+, +), quadrant 2 is (−, +), quadrant 3 is (−, −), quadrant 4 is (+, −).
- Look at the **signs** of the two numbers to decide.
- A point with a **0** in it sits **on an axis**, not in any quadrant.

State the quadrant, or the axis, each point lies on.



1. (3, 4)

2. (−4, −2)

3. (−2, 5)

Answer: _____

Answer: _____

Answer: _____

4. (6, −1)

5. (0, 3)

6. (−5, 0)

Answer: _____

Answer: _____

Answer: _____

State the quadrant, or the axis, each point lies on.

7. (−8, −8)

8. (0, −6)

9. (9, 7)

Answer: _____

Answer: _____

Answer: _____

Worksheet 3: Quadrants of the number plane

Week 2 · Worksheet 3 of 6 · continued

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MEDIUM

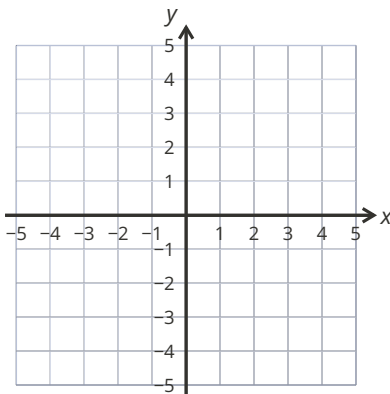
6 QUESTIONS

WORKING THE OTHER WAY

- Asked for a point in a quadrant? Choose the signs first, then pick any numbers.
- Reflecting in the y-axis flips the sign of x; in the x-axis it flips the sign of y.

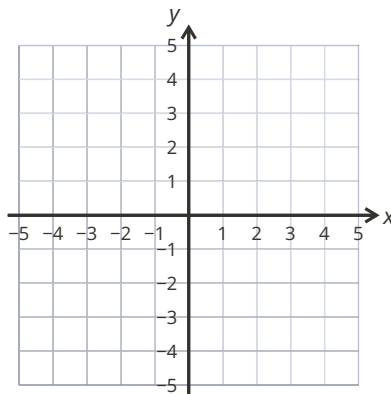
Plot the point, then state which quadrant it is in.

10. A(-3, 2)



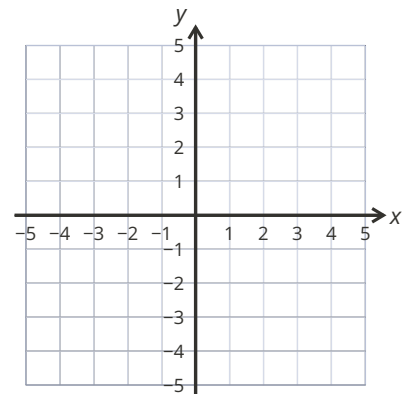
Answer: _____

11. B(4, -3)



Answer: _____

12. C(-2, -4)



Answer: _____

Answer each question.

13. In which quadrant is a point with negative x and positive y?

Answer: _____

14. Give a point in quadrant 4 with a y-value of -3.

Answer: _____

15. A(2, 5) is reflected in the y-axis. Which quadrant is the image in?

Answer: _____

CHALLENGING

3 QUESTIONS

BEFORE YOU START

- Sketch the four quadrants and label them before you decide.

Read each one, draw your own diagram, then answer the question.

16. Sketch a number plane and label all four quadrants. Then mark one point in each quadrant and write its coordinates.

Draw your diagram here

Answer: _____

17. Which quadrant contains every point where x is positive and y is negative? Explain how you know.

Draw your diagram here

Answer: _____

18. Explain why the point (0, 0) is not in any quadrant.

Draw your diagram here

Answer: _____

Worksheet 4: Graphing tables of values

Week 2 · Worksheet 4 of 6 · 18 questions

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Every column of a table of values gives one point (x, y). Plot the points, then rule a line through them.

EASY

9 QUESTIONS

FROM TABLE TO POINTS

- Take the top and bottom number of a column together: that is one point.
- x: 1, 2, 3 y: 4, 5, 6 gives the points (1, 4), (2, 5) and (3, 6).
- Plot every column, then rule **one straight line** through them all.

Write the points from each table as coordinate pairs.

1. Write the three points.

x	1	2	3
y	4	5	6

Answer: _____

2. Write the three points.

x	0	1	2
y	0	2	4

Answer: _____

3. Write the three points.

x	1	2	3
y	3	6	9

Answer: _____

4. Write the three points.

x	-1	0	1
y	1	2	3

Answer: _____

5. Write the three points.

x	0	1	2
y	5	4	3

Answer: _____

6. Write the three points.

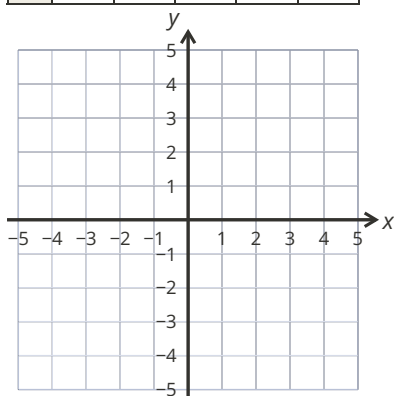
x	-2	0	2
y	-4	0	4

Answer: _____

Plot the points from the table, then rule the line.

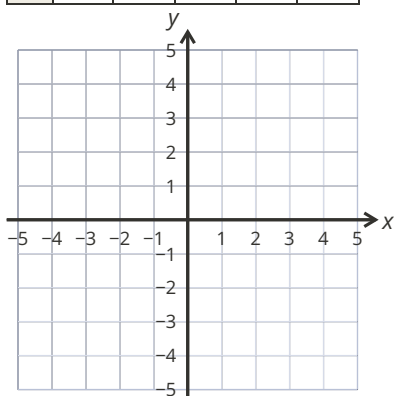
7. Plot and rule the line.

x	-2	-1	0	1	2
y	-1	0	1	2	3



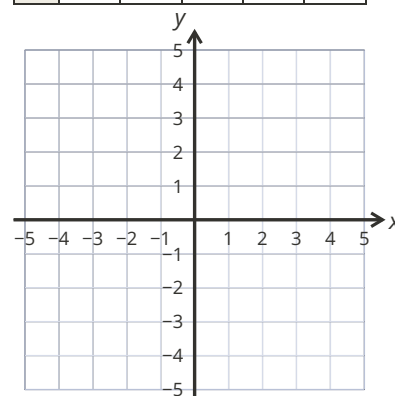
8. Plot and rule the line.

x	-2	-1	0	1	2
y	-4	-2	0	2	4



9. Plot and rule the line.

x	0	1	2	3	4
y	5	4	3	2	1



Worksheet 4: Graphing tables of values

Week 2 · Worksheet 4 of 6 · continued

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6 QUESTIONS

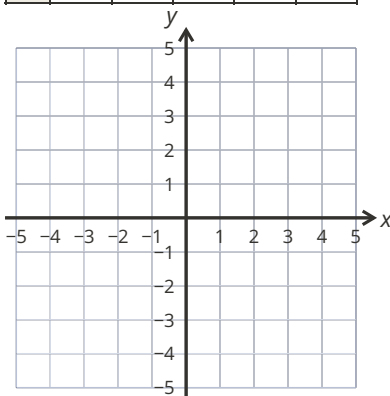
GRAPHING NEATLY

- Use a ruler and let the line run right across the grid, past the end points.
- If one point is out of line with the others, check that column again.

Plot the points from the table, then rule the line.

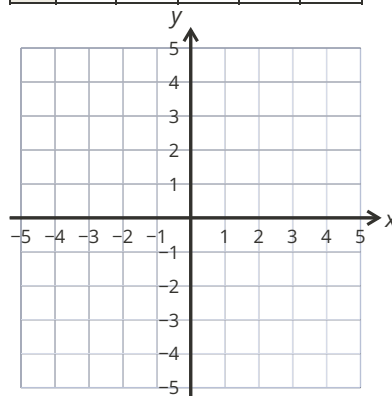
10. Plot and rule the line.

x	-2	-1	0	1	2
y	-3	-2	-1	0	1



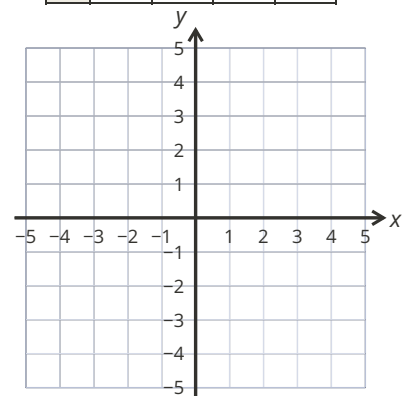
11. Plot and rule the line.

x	-2	-1	0	1	2
y	2	1	0	-1	-2



12. Plot and rule the line.

x	-1	0	1	2
y	-3	0	3	6



Answer each question.

13. A table gives the points (0, 1), (1, 2) and (2, 3). Do they lie in a straight line?

Answer:

14. Which point in this list is not in line with the others: (0, 0), (1, 2), (2, 4), (3, 5)?

Answer:

15. A line passes through (0, 2) and (1, 3). Write one more point on the line.

Answer:

CHALLENGING

3 QUESTIONS

BEFORE YOU START

- Draw and number both axes first, then plot the points from the table.

Read each one, draw your own diagram, then answer the question.

16. Draw a number plane from -4 to 4 on both axes. Plot the points (-2, -1), (0, 1), (2, 3) and rule the line through them.

Draw your diagram here

Answer:

17. Plot the points (0, 4), (1, 3), (2, 2) and (3, 1), rule the line, then use your graph to find y when x = 4.

Draw your diagram here

Answer:

18. Plot the points (0, 0), (1, 2), (2, 4) and (3, 5). One of them is not in line with the others. Which one, and how can you tell from the graph?

Draw your diagram here

Answer:

Worksheet 5: Completing tables of values

Week 3 · Worksheet 5 of 6 · 21 questions

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Substitute each x-value into the rule to work out its y-value.

EASY

12 QUESTIONS

FILLING IN A TABLE

- Replace x with the number from the top row, then work the rule out.
- For $y = 2x + 1$ with $x = 3$: $y = 2 \times 3 + 1 = 7$.
- Work one column at a time and write the answer underneath.

Complete each table of values.

1. $y = x + 3$

x	0	1	2	3	4
y					

2. $y = 2x$

x	0	1	2	3	4
y					

3. $y = x - 2$

x	0	1	2	3	4
y					

4. $y = 5x$

x	0	1	2	3	4
y					

5. $y = x + 6$

x	0	1	2	3	4
y					

6. $y = 10 - x$

x	0	1	2	3	4
y					

7. $y = 4x$

x	0	1	2	3	4
y					

8. $y = x + 8$

x	0	1	2	3	4
y					

9. $y = 8 - x$

x	0	1	2	3	4
y					

Complete the table, then write the points as coordinate pairs.

10. $y = x + 1$

x	-2	-1	0	1	2
y					

11. $y = 3x$

x	-2	-1	0	1	2
y					

12. $y = 2x + 1$

x	-1	0	1	2	3
y					

Answer: _____

Answer: _____

Answer: _____

Worksheet 5: Completing tables of values

Week 3 · Worksheet 5 of 6 · continued

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MEDIUM

6 QUESTIONS

CAREFUL WITH NEGATIVES

- Multiply before you add or subtract: for $y = 3x - 2$ with $x = -1$, $3 \times (-1) = -3$, then $-3 - 2 = -5$.
- A rule like $y = 10 - x$ means take x away from 10.

Complete each table of values.

13. $y = 3x - 2$

x	-2	-1	0	1	2
y					

Answer:

14. $y = 4 - x$

x	-2	-1	0	1	2
y					

Answer:

15. $y = 2x + 5$

x	-1	0	1	2	3
y					

Answer:

16. $y = 10 - 2x$

x	-2	-1	0	1	2
y					

Answer:

17. $y = 5x - 1$

x	-1	0	1	2	3
y					

Answer:

18. $y = x - 4$

x	-2	-1	0	1	2
y					

Answer:

CHALLENGING

3 QUESTIONS

BEFORE YOU START

- Draw the table first with x on top and y underneath, then fill it in.

Read each one, draw your own diagram, then answer the question.

19. Make a table of values for $y = 2x - 3$ using $x = -2$ to 2. Set it out with x on top and y underneath.

20. Make a table of values for $y = 12 - 3x$ using $x = 0$ to 4, then write down the point where y is 0.

21. For $y = 4x$, which x -value gives $y = 20$? Show how you worked it out.

Draw your diagram here

Draw your diagram here

Draw your diagram here

Answer:

Answer:

Answer:

Worksheet 6: Graphing linear equations

Week 3 · Worksheet 6 of 6 · 21 questions

NSW syllabus outcome

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Complete a table of values for the equation, plot the points, then rule the line.

EASY

12 QUESTIONS

GRAPHING AN EQUATION

- Complete the table first, one column at a time.
- Each column gives a point (x, y). Plot them all.
- Rule **one straight line** through the points, right across the grid.

Complete each table of values for the equation.

1. $y = x + 2$

x	-2	-1	0	1	2
y					

2. $y = 2x$

x	-2	-1	0	1	2
y					

3. $y = x - 1$

x	-2	-1	0	1	2
y					

4. $y = 3x$

x	-1	0	1	2	3
y					

5. $y = 2x + 1$

x	-2	-1	0	1	2
y					

6. $y = 5 - x$

x	0	1	2	3	4
y					

7. $y = x + 4$

x	-2	-1	0	1	2
y					

8. $y = 4x$

x	-1	0	1	2	3
y					

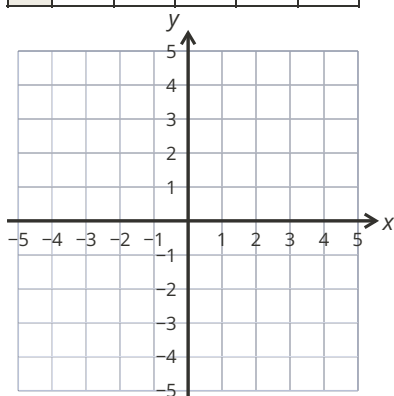
9. $y = 3x + 2$

x	-2	-1	0	1	2
y					

Complete the table, then plot the points and rule the line.

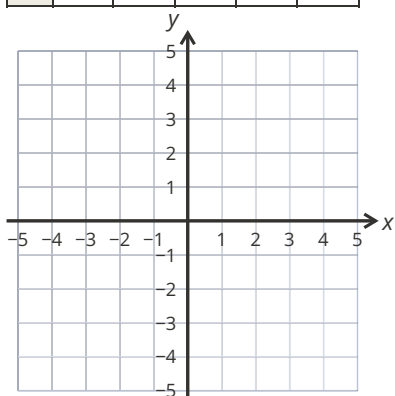
10. $y = x + 2$

x	-2	-1	0	1	2
y					



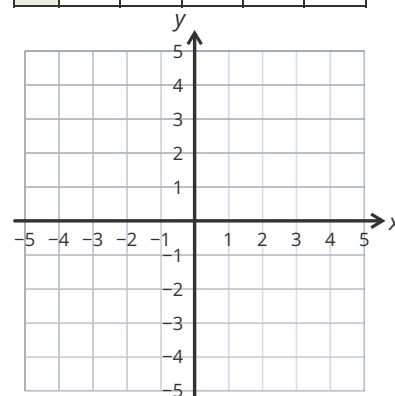
11. $y = 2x$

x	-2	-1	0	1	2
y					



12. $y = 5 - x$

x	0	1	2	3	4
y					



Worksheet 6: Graphing linear equations

Week 3 · Worksheet 6 of 6 · continued

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6 QUESTIONS

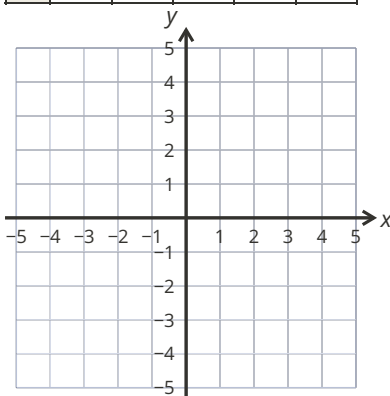
CHECKING YOUR GRAPH

- All the points should sit in a straight line. If one does not, check that column.
- Substitute a point back into the equation to check it works.

Complete the table, then plot the points and rule the line.

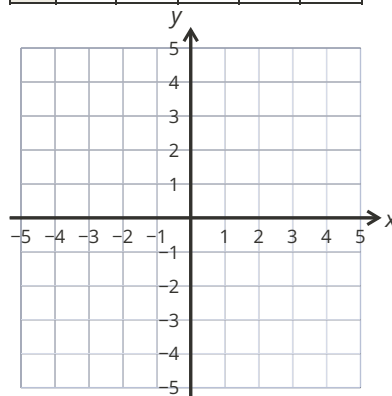
13. $y = x - 1$

x	-2	-1	0	1	2
y					



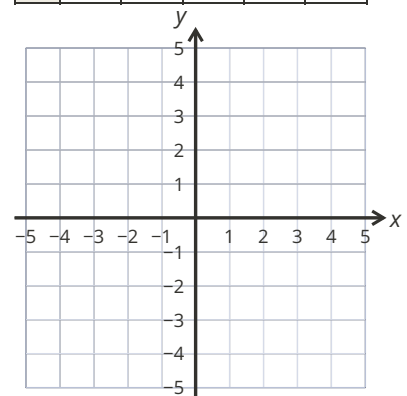
14. $y = 3x$

x	-2	-1	0	1	2
y					



15. $y = 2x - 1$

x	-1	0	1	2	3
y					



Answer each question.

16. Does the point (2, 5) lie on the line $y = 2x + 1$? Show your working.

Answer: _____

17. Does the line $y = 3x$ pass through the point where the axes cross?

Answer: _____

18. For $y = x + 4$, what is y when $x = 6$?

Answer: _____

CHALLENGING

3 QUESTIONS

BEFORE YOU START

- Draw and number both axes, make your table, then plot and rule the line.

Read each one, draw your own diagram, then answer the question.

19. Make a table of values for $y = x + 1$ using $x = -2$ to 2, then draw its graph on a number plane.

Draw your diagram here

Answer: _____

20. Make a table of values for $y = 2x - 2$ using $x = -1$ to 3, then draw its graph and write down where it crosses the y -axis.

Draw your diagram here

Answer: _____

21. Graph $y = 4 - x$ for $x = 0$ to 4, then use your graph to find y when $x = 1.5$.

Draw your diagram here

Answer: _____